
Computergestützte Inferenz und Analyse von PRDM9-unabhängigen meiotischen Rekombinations-Hotspots in Populationen der Europäischen Mönchsgrasmücke *Sylvia atricapilla*

Computational inference and analysis of prdm9 independent meiotic
recombination hotspots in European Blackcap *Sylvia atricapilla* populations

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Contents

1	Abstract	2
2	Introduction	2
2.1	Overview of Meiosis	2
2.2	Meiotic Recombination Hotspots and <i>prdm9</i>	3
2.3	Recombination occurs even in the absence of <i>prdm9</i>	3
2.4	Methods for Studying Meiotic Recombination Hotspots	4
2.5	Main objectives	4
3	Materials & Methods	5
3.1	Data Acquisition	5
3.2	Population Genomics Statistics	6
3.3	Inference of Recombination Rate Maps	6
3.4	Analysis of Recombination Rate Landscape Over The Blackcap Genome	7
3.5	Inference of Meiotic Recombination Hotspots	8
4	Results	9
4.1	Population genomics statistics of the european blackcap metapopulation	9
4.2	Genomic Features of the blackcap metapopulation	11
4.3	Inference of meiotic recombination maps	12
4.4	Wavelet transformation identifies meiotic recombination hotspots in blackcaps	15
5	Discussion	18
5.1	Recent population expansion and high diversity in the blackcap genome	18
5.2	Strong correlation between gc content and recombination rate in blackcaps	19
5.3	Blackcap 5' UTRs are enriched in recombination hotspots	19
5.4	Conclusions	21
6	References	22
7	Declaration	31

1 Abstract

Ever since its discovery, the role of *prdm9* has been described as detrimental to successful propagation of meiotic recombination in sexually reproducing eukaryotes. *Prdm9*'s key role is the trimethylating of histone 3 lysine 4 and histone 3 lysine 36, which indicate the locations where recombination should take place, thus moving the so called hotspots away from functional genome elements, such as genes and promoters. Recently studies have uncovered that not all sexually reproducing organisms carry the gene, yet still undergo recombination successfully, most notably *Aves*, *Crocodylia*, *Canes*, & *Percomorpha*. Moreover, it was discovered that *prdm9* independent recombination hotspots occur even in species carrying the gene. This finding underscores the importance of understanding the underlying mechanism of *prdm9* independent recombination. In this study I seek to computationally investigate this alternate pathway of recombination by computing the population scaled recombination rate of *Sylvia atricapilla* (European blackcap), investigating the recombination landscape and identifying hotspots from recombination rate map. To do so I utilize *pyrho*, a software able to compute demography informed population scaled recombination rate as well as continuous wavelet transformation which I use to call hotspot peaks from local background noise via a peak prominence threshold, ultimately detecting 712 hotspots when using a threshold of 90% percentile of recombination values and 6142 hotspots when using a threshold of 30% percentile of recombination values

2 Introduction

2.1 Overview of Meiosis

Meiosis is a specialized cell division process that generates haploid gametes from diploid germ cells in sexually reproducing organisms (Handel and Schimenti 2010). Following a single round of DNA replication, two sequential divisions take place, during which homologous chromosomes segregate. A key feature of meiosis is **meiotic recombination**, which takes place during Prophase I of meiosis and reshuffles alleles between homologous (parental) chromosomes, ultimately enhancing genetic diversity (Hunter 2015; Baudat, Imai, and Massy 2013). Recombination is a complicated process, involving the induction of programmed DNA double-strand breaks (DSBs), 5' strand resection, homologous strand invasion, Holliday junction formation, and

resolution into crossover (CO) or non-crossover (NCO) events, including DSBs repair (Lange et al. 2016; Börner, Kleckner, and Hunter 2004). These events occur at specific chromosomal loci termed **recombination hotspots**. Recombination hotspots are essential for meiosis, as at least one recombination event has to occur per chromosome in order for gametogenesis to be successful (Lange et al. 2016).

2.2 Meiotic Recombination Hotspots and *prdm9*

In most mammals, the location of recombination hotspots is largely determined by **PRDM9** (Powers et al. 2016; Imai et al. 2017; Cavassim et al. 2022; AbuAlia et al. 2024). PRDM9 is a histone methyl-transferase consisting of three domains: 1. The KRAB domain that recruits the recombination machinery; 2. The PR/SET domain that catalyzes the trimethylation of H3K4 and H3K36 hence marking the hotspot locations; 3. The hypervariable C_2H_2 zinc finger array that binds to specific DNA motifs (Powers et al. 2016; Imai et al. 2017; Cavassim et al. 2022). Once bound to its DNA motifs, PRDM9 trimethylates H3K4 and H3K36; recruits factors such as SPO11, MLH1, DMC1/RAD51, and ZCWPW1/2 to initiate double-strand breaks, promote homologous strand exchange, and ensure DSBs repair (Parvanov et al. 2017; Tian, Billings, and Petkov 2021). However, with every meiotic event, PRDM9 imposes evolutionary pressure on itself: with each repair of DSBs at its DNA binding sites, *prdm9* motif mutations can accumulate, gradually eroding motifs and forcing a continuous turnover of PRDM9's zinc finger array (Baker et al. 2015).

2.3 Recombination occurs even in the absence of *prdm9*

While the presence of the *prdm9* gene is a must in humans, mice, apes etc.; some organisms such as birds undergo meiotic recombination even the absence of *prdm9* (Cavassim et al. 2022; Pigozzi 2016). In birds the *prdm9* gene has been lost during evolution, yet their chromosomes undergo meiotic recombination seamlessly (Pigozzi 2016). Furthermore, a recent study (Joseph et al. 2024) showed the prevalence of *prdm9*-independent recombination hotspots even in organisms that carry the *prdm9* gene. These findings raise relevant questions such as: 1. Why do organisms carrying *prdm9* retain *prdm9*-independent recombination hotspots? 2. Do any cross-over/non-crossover events take place in *prdm9*-independent recombination hotspots? 3. What are the underlying recombination mechanisms in *prdm9*-independent hotspots? 4. Are these *prdm9*-independent hotspots evolutionary conserved and if so why?

2.4 Methods for Studying Meiotic Recombination Hotspots

The traditional methodology for studying meiotic recombination involves the use of cytological methods, conducted on spermatocytes or oocytes. The methods include immuno-staining of meiotic markers (such as PRDM9, MLH1, SYCP3, SYCP1, HORMAD1/2) as well as Histone Modifications (such as H3K4me3/36) (J. Lee et al. 2019; Parvanov et al. 2016; Lam et al. 2019; Hinch et al. 2020; Pigozzi 2016). While the immuno-staining methods provide a good broad-scale overview of meiosis it also has several drawbacks. They lack of resolution, are unable to quantify the strength of a recombination event, have limited ability to determine the evolutionary history of recombination events across the genome, as well as difficulty to establish a baseline as the recombination landscape if meiosis factors differ in different species, such as the presence/absence of *prdm9* (Pigozzi 2016; Baudat, Imai, and Massy 2013). However, there is another angle at which meiotic recombination can be studied. Computational methods utilizing linkage disequilibrium, that is the degree of non-random association between two alleles, are a good alternative for determining the location of recombination events along the genome (Johnston 2024; G. A. T. McVean 2002). LDhat estimates the composite likelihood of recombination rates being observed at a genomic position by modeling recombination rates under the assumption of constant population sizes; LDhot is a extension of LDhat which determines the presence of recombination hotspots given a recombination rate via a likelihood ratio test and LDhelmet which is a re-implementation of LDhat that makes it possible to analyze highly divergent species (Montgomery Slatkin 2008; G. McVean, Awadalla, and Fearnhead 2002; Auton and G. McVean 2007; Chan, Jenkins, and Song 2012). While these tools yield fine scale resolution of recombination hotspots, they neither account for the complicated demographic history such as bottlenecks, recent expansions or population structure nor are they computationally feasible for large scale population studies (Spence and Song 2019). Pyrho solves this issue by taking advantage of two locus sample likelihood and computes recombination rates by maximizing a composite likelihood built from two-SNP haplotype probabilities, where those probabilities are derived under a demography-aware coalescent model, such as one simulated through msprime (Spence and Song 2019; Baumdicker et al. 2022).

2.5 Main objectives

In this study I seek investigate the fine scale *prdm9*-independent meiotic recombination landscape. To this end, I will extend the analysis of the recombination rate landscape previously

conducted on 30 European blackcap (*Sylvia atricapilla*) individuals to a larger dataset of 100 individuals (Bascón-Cardozo et al. 2024). European Blackcaps are an excellent choice of model organism as there are extensive genome sequences, demography models and previous studies which have been conducted on blackcaps, albeit at a lower scale (Ishigohoka and Liedvogel 2025; Bascón-Cardozo et al. 2024; Bours et al. 2023). The population scaled recombination rate landscape will be estimated through `pyrho` using previously published demography models via `msprime` and individual genetic variation data (Ishigohoka and Liedvogel 2025; Bascón-Cardozo et al. 2024). Following this the distribution of the recombination rate along the landscape of European blackcaps, overlap with genomic functional elements, GC content, genes, intergenic regions, UTRs, exons, codons, Nucleotide π , SNP count, Tajimas D (Bascón-Cardozo et al. 2024; Singhal et al. 2015). Lastly, to identify the most likely recombination hotspots in the blackcap genome, I will use an alternative to the sliding-window approach of Wooldridge and Dumont 2023. Rather than averaging recombination rates over large sliding windows and losing resolution as well as risk flattening peaks, I will utilize continuous wavelet transformation which circumvent these issues (González-Nuevo et al. 2006) to identify peaks that persist across several genomic windows starting at small 50 bp and going up to 10000 bp.

3 Materials & Methods

3.1 Data Acquisition

To computationally identify and characterize `prdm9`-independent hotspots in European blackcap (*Sylvia atricapilla*), hereafter referred to as blackcap, on 15 March 2025 I retrieved:

- Population level blackcap whole genome variant call VCF, blackcap individual identifiers and inferred demography history data through SMC from (Ishigohoka and Liedvogel 2025).
- Blackcap genome assembly, GFF and genome sizes from NCBI with genome accession number GCF_009819655.1 (Sayers et al. 2024).
- Zebra finch to blackcap lift-over chain file `taeGut2ToGCF_009819655.1.over.chain` from the UCSC genome browser (Perez et al. 2024; Hinrichs et al. 2006).
- Zebra finch recombination rate maps from Singhal et al. 2015

As the demography data from Ishigohoka and Liedvogel 2025 was computed for each blackcap sub population, I merged the msmc2 table by taking the average of each input across all tables. I automated this process with a custom script `merge_msmc2_table.py`.

3.2 Population Genomics Statistics

To describe the different aspects of genomic variation, such as average pairwise sequence variation between individuals, frequency of polymorphic sites and deviation from natural evolution I computed key population genomics statistics: **Tajima's D**, **Window π** and **SNP density** using VCFTools (v0.1.16) (Danecek et al. 2011; Tajima 1989; Nei and Li 1979). To compute the statistics I first sampled 100 random individual identifiers using GNU `shuf` and saved them into a file called `n100_blackcap.panel`. Using this file, I then extracted the corresponding SNP records with a minor allele frequency maf of 0.05% and computed the statistics in window sizes of 1000, 5000, 10,000 and 100,000 base pairs (bp). Next, I plotted the genome-wide distribution of each statistic against their midpoints, which were calculated through the formula: $\frac{Start+End}{2}$. Lastly, divided the statistics into three equal quantile-based bins representing "low", "medium" and "high" values and visualized the proportion of each quantity per chromosome. I automated the visualization with a custom script called `plot_popgen_stats.py`.

3.3 Inference of Recombination Rate Maps

To compute the population scaled recombination rate maps in blackcaps, I utilized a demography-aware method to infer fine-scale recombination rates between SNPs called `pyrho` (Spence and Song 2019). I first precomputed the likelihood tables using `pyrho make_table` needed for estimating the population-specific recombination rate (Spence and Song 2019). The table was precomputed with the following parameters:

- Mutation rate μ : 4.6×10^{-9} , as reported in Bascón-Cardozo et al. 2024
- Decimal relative tolerance: 0.1 as reported in Bascón-Cardozo et al. 2024
- Population size (i.e., number of haplotypes): 200
- Moran population size: 200
- The averaged MSMC2 demography history data.

I automated this process with a custom script, `make_table.sh`. Secondly, I optimized the `block_penalty` and `window_size`, two hyperparameters' required for the computation of the population-scaled recombination rate, using `pyrho` hyperparam (Spence and Song 2019). I ran 10 simulations over the following grid of values:

- **Block Penalty:** 10, 20, 30, 40, 50
- **Window Size:** 25, 30, 50, 100, 1000

The optimization used the Averaged MSMC2 demography history and precomputed likelihood table from the previous step. This step was automated using `optimize_hyperparameters.sh`. `pyrho` hyperparam returns a table with the pearson and spearman correlation coefficients and the $L2$ norm. I selected a **window size of 100** with a **block penalty of 20** as they maximized the pearson and spearman correlation coefficient while minimizing the $L2$ norm, as specified by (Spence and Song 2019). I sampled the same 100 individuals from the VCF data as in section 2.2, but only with allele count of at least 2. Lastly, using the precomputed likelihood, sampled individual SNP data table, window size of 100 with a block penalty of 20 I computed population-scaled the recombination rate with `pyrho` `optimize` (Spence and Song 2019). This process was automated with a custom script called `pyrho_optimize.sh`

3.4 Analysis of Recombination Rate Landscape Over The Blackcap Genome

To investigate the population scaled recombination rate landscape in blackcaps, I converted the raw recombination rate to cM/Mb by averaging over genomic windows of 10 kilobases and multiplying by 10^8 . Then, I calculated the midpoint of the genomic coordinates by the formula: $\frac{Start+End}{2}$. Lastly, I plotted the recombination rate in cM/Mb over the genomic midpoint coordinates. I investigated the distribution of recombination rate strength per chromosome. To do this, I sorted the recombination rate per chromosome by signal strength and divided it into equal tertiles ("low", "medium" and "high"). I visualized the data on a chromosomal basis using pie charts.

To gain a more indepth understanding of the recombination landscape I incorporated the gene density and gc content into the analysis. I extracted the genes from the blackcap gff file using a custom script `clean_raw_gff.py` and computed the gene density over genomic windows and divided by 1 Mb to get $\frac{genes}{Mb}$. I generated genomic bins of size 100000 using `PyRanges`, counted the number of genes overlapping with each window and divided by the window size and

plotted the distribution of the gene density along the genome and binned the gene density into tertiles representing "low", "medium" and "high". To compute the gc content, I used `bedtools makewindows` to make genomic windows per each chromosome, `bedtools nuc` to compute the gc content over the windows, which I automated with the script `compute_gc_content.sh`. Next, I plotted the gc content over each chromosome and binned it into three tertiles ("low", "medium" and "high") values to visualize how the gc content is distributed over the genome. I automated the plotting in the steps above using the script `plot_popgen_stats.py`.

To quantify the correlation between the population genomic statistics, gc content, gene density and the recombination rate I first intersected each feature with one another using `pyRanges`, computed the pearson correlation coefficient between each genomic feature using `SciPy` and visualized the results in a heatmap (Virtanen et al. 2020). I automated this analysis with a custom script called `compute_feature_correlation.py`.

Lastly, I investigated the average profile of the recombination rate over genomic features (UTR's, CDS, Exons, Intergenic Regions and Genes), which I extracted from the blackcap GFF using the `clean_raw_gff.py`. I used `deeptool's computeMatrix` to calculate the average profile of the recombination rate over these regions and `plotHeatmap` to plot the average profile and heatmap. The average recombination rate profile was computed for ± 20 kb and zeros were not reported. The script for computing the matrix and plotting the profile are: `computeRhoMatrix.sh` and `plotHeatmap.sh`, respectively.

3.5 Inference of Meiotic Recombination Hotspots

To call recombination hotspots from recombination rate maps, I applied **Continuous Wavelet Transform (CWT)** to detect multi-scale peaks in the signal. To do so the wavelet is continuously rescaled across a grid of specified scales and shifted along the signal, producing coefficients that quantify local similarity between the signal and the wavelet shape at each position and scale. The higher the similarity, the higher coefficients (Singh, Rawat, and Raghuthaman 2022; González-Nuevo et al. 2006). Before computing the cwt with `PyWavelet`, I smoothed the recombination rate ρ signal using `SciPy's` one dimensional gaussian filter (`gaussian_filter1d`) ($\sigma = 5$) to remove excessive high frequency noise. Then, I generated 200 logarithmically spaced scales between 50 and 600. With an recombination rate genomic range of 22 bp this corresponds to

genomic bin size between ~ 1.1 kb to ~ 13.2 kb. After computing the wavelet coefficients, I normalized the coefficients at each scale independently through z-score. Following this, I selected only those coefficients, which had a z-score: $z > 2.0$ and persisted in atleast 30% of the scales. The peaks were then identified using SciPy's `find_peaks` with a minimum peak prominence of at least the 90th percentile. This process was automated using the script `find_hotspots.py`. Lastly, I used the package `cobind.py` to quantify the amount of overlap between the identified recombination hotspots and genomic features (UTRs', codons, genes, intergenic regions) using the npmi metric which gives the enrichment of overlaps relative to what is expected at random (Ma et al. 2023)

To test if the wavelet transform methodology is also applicable to other prdm9 independent recombining species I ran the same analysis on LDHelmet recombination maps generated by Singhal et al. 2015.

4 Results

4.1 Population genomics statistics of the european blackcap metapopulation

To investigate the deviation from neutral evolution, average pairwise nucleotide diversity and snp density I computed three population genomics statistics: tajima D , diversity π and SNP density across the autosomal chromosomes on a blackcap metapopulation consisting of 100 individuals. Tajima's D showed two types of patterns across each of the 33 blackcap autosomal chromosomes (Figure 1, Supplementary Figure 1). The large chromosomes one through 12, showed relatively stable negative tajima D values across the chromosomal body, with the exception of one peak in chromosomes 2 and 4 at $\sim 20Mb$ with a tajima D value of 0 and a sharp valley with a tajima D value of over -2 at $\sim 7Mb$ on chromosome 6, and the end of chromosome 12 that contained several peaks of positive tajima D (Supplementary Figure 1). The chromosomal ends of these macrochromosomes had negative tajima D values which were close to 0. The blackcap microchromosomes displayed opposite patterns, the subtelomeric regions had valleys beyond values of tajima $D : -2$ while the chromosomal body showed stable yet tiny fluctuations in the tajima D values, which were negative yet closer to 0 (Supplementary Figure 1).

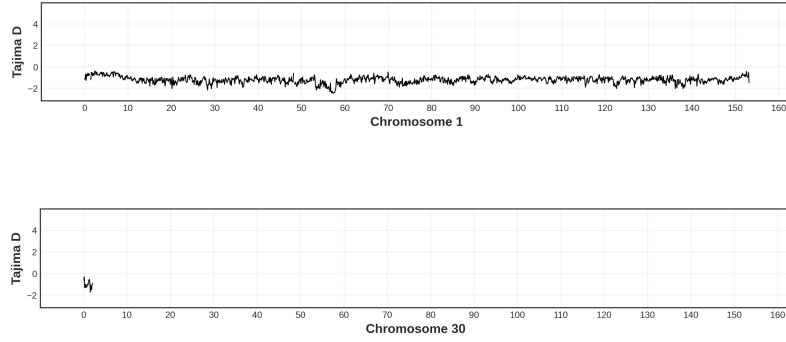


Figure 1. Tajima's D across chromosomes 1, 10, and 30 in 100 kb windows for a European Blackcap metapopulation (100 individuals from continental and island populations).

Compared to tajima's D , the SNP density and nucleotide diversity π displayed more visible patterns across the chromosomes (Figure 2). The macrochromosomes had an average SNP density and nucleotide diversity π of 50 and 0.005 over 100 Mb window bins, respectively (Figure 2; Supplementary Figure 2 3). In addition to this, both values displayed regularly interspaced sharp valleys occurring at ~ 10 Mb. Chromosome 1, in contrast to the other, macrochromosomes had a sharp peak in its chromosomal end (Supplementary Figures 2 3). The microchromosomes, on the other hand, revealed stable patterns, with reduced signal in the subtelomeric regions with occasional valleys in SNP density and diversity π along the chromosomal axis (Supplementary Figures 23). Here again, microchromosomes displayed overall SNP density and diversity π across the chromosomal axis relative to the macrochromosomes. Interestingly the trends of SNP density, diversity π and tajima D have a medium positive correlation (pearson correlation r ; Figure 7) with each other and decreases in the signal of these metrics are observed to occur in similar genomic positions (Figure 12, Supplementary Figures 13) other,

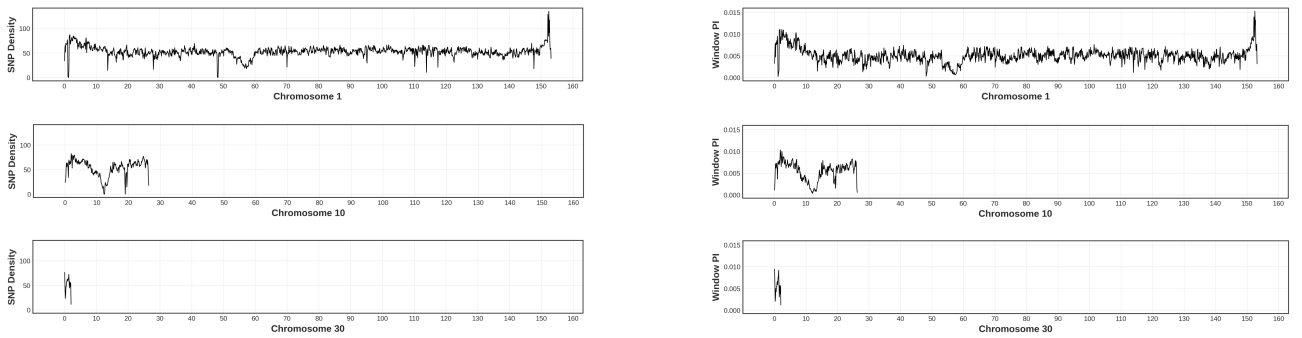


Figure 2. SNP density (left) and nucleotide diversity (π) (right) across chromosomes 1, 10, and 30 in 100 kb windows.

4.2 Genomic Features of the blackcap metapopulation

To investigate the genomic features of the blackcap metapopulation as well as their relationship to the recombination landscape of the species, I computed the GC content and the gene density across the blackcap genome in 100 kb windows. The blackcap genome has an average GC content of 41.8% across all autosomal chromosomes (Table 1, Figure 3, Supplementary Figure 4). Overall the GC content has two types of patterns across the blackcap chromosomes. On chromosomes 1-12, it shows elevated GC content in the subtelomeric regions, with several valleys in the chromosomal body corresponding with low GC content (Figure 3, Supplementary Figure 4). The microchromosomes, contained another pattern. They showed elevated GC content over all the chromosomal body with GC content trends being over 50% and relatively stable (Supplementary Figure 4). Furthermore, this trend in GC content values with decreasing chromosome size was even more pronounced when the values were binned into equal tertiles (Figure 3; Supplementary Figure 4). Here, the microchromosomes had a larger proportion of high GC content as opposed to the macrochromosomes.

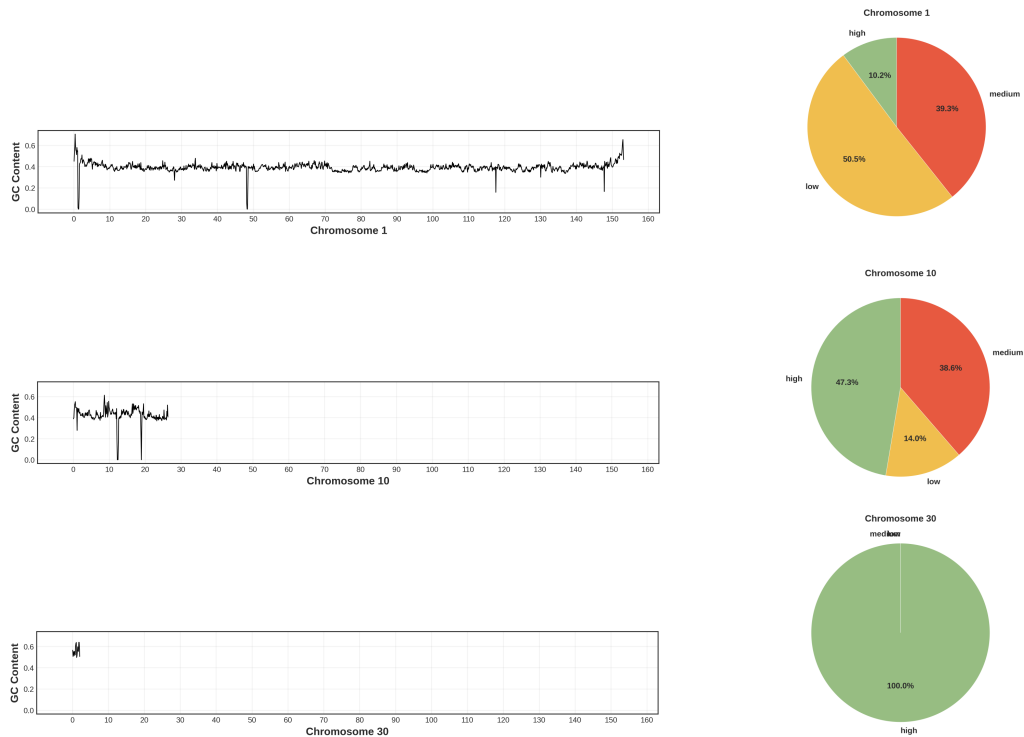


Figure 3. (Left) GC content across chromosomes 1, 10, and 30 in 100 kb windows and (Right) GC content binned into equal tertiles (*low*, *medium*, *high*).

The blackcap genome has an average gene density of 33.83 $\frac{genes}{Mb}$ (Table 1). The macrochromo-

mosomes show low gene density across their length, with elevated gene density in the subtelomeric regions (Figure 4, Supplementary Figure 5) and have average gene density lower than microchromosomes (Table 1). Microchromosomes, on the other hand, have an higher average gene density along their length and lower gene density in their subtelomeric regions compared to the macrochromosomes (Table 1, Figure 4, Supplementary Figure 5). Furthermore, the GC content and Gene Density showed medium positive correlation (pearson's $r \approx 0.6$, Figure 6), while the Gene density had a weak correlation, with tajima's D , diversity π and SNP density ($r \approx 0.2$; Figure 6).

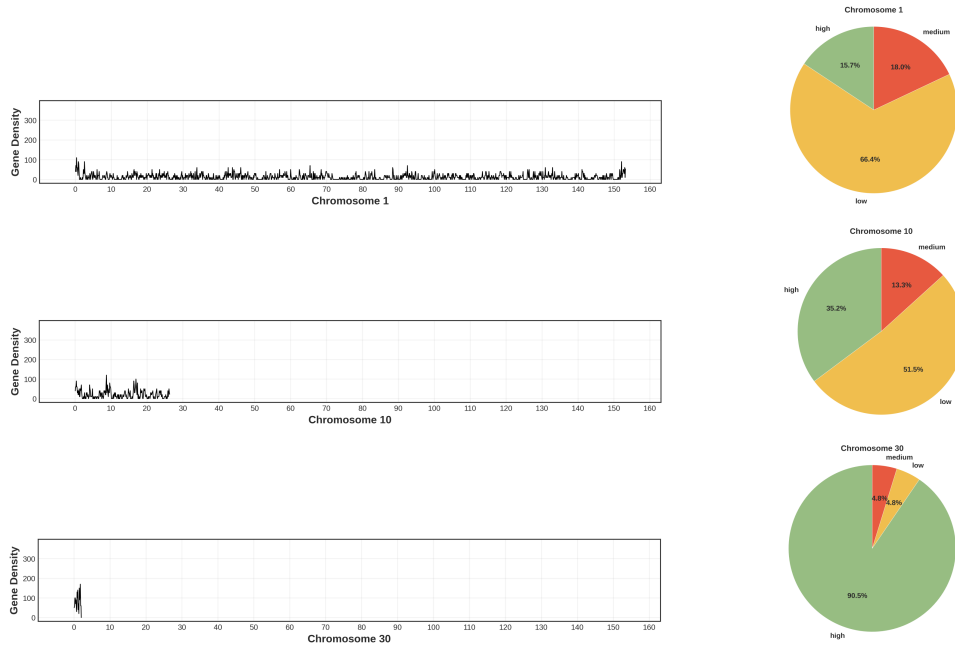


Figure 4. (Left): Gene density across chromosomes 1, 10, and 30 in 100 kb windows. **(Right):** Gene density binned into equal tertiles (*low*, *medium*, *high*).

4.3 Inference of meiotic recombination maps

To investigate the recombination landscape in european blackcaps and gain insights in the prdm9 independent recombination that happens in organisms lacking prdm9 I computed meiotic recombination rate maps using pyrho (Joseph et al. 2024; Cavassim et al. 2022; Spence and Song 2019). The recombination landscape on the blackcap population has a genome wide average of $0.48 \frac{cM}{Mb}$ (Table 1). Most of the high recombination rate signal is concentrated in the subtelomeric regions, as reported for other bird species (zebra finches and long tailed finches) and contained several spikes in recombination length over the chromosomal axis. (Singhal et al. 2015). The recombination rate average per chromosome, seems to vary with chromosomal length, the mi-

crochromosomes had a higher proportion of high recombination rate regions and a higher recombination rate average then the macrochromosomes (Table 1, Figure 5, Supplementary Figure 6).

To compare the recombination rate landscape with other genomic features I computed it's pearson correlation with the features mentioned above. The recombination rate seemed to have a medium positive correlation with the GC Content and window π ($r \approx 0.5$), Gene density and SNP Density ($r \approx 0.4$) and a weak correlation with tajima's D ($r \approx 0.25$). Next, to visualize the recombination landscape over functional genomic regions I computed the average profile and enrichment of the recombination rate. The recombination rate showed a narrow high peak on 5' UTRs with an average value of $\rho = 2.5 \times 10^{-7}$, small peaks on CDS and exons with an average $\rho = 1.3 \times 10^{-7}$ a flat background across genes and 3' UTRs' with $\rho = 1.3 \times 10^{-7}$ and no signal on intergenic regions (Figure 7)

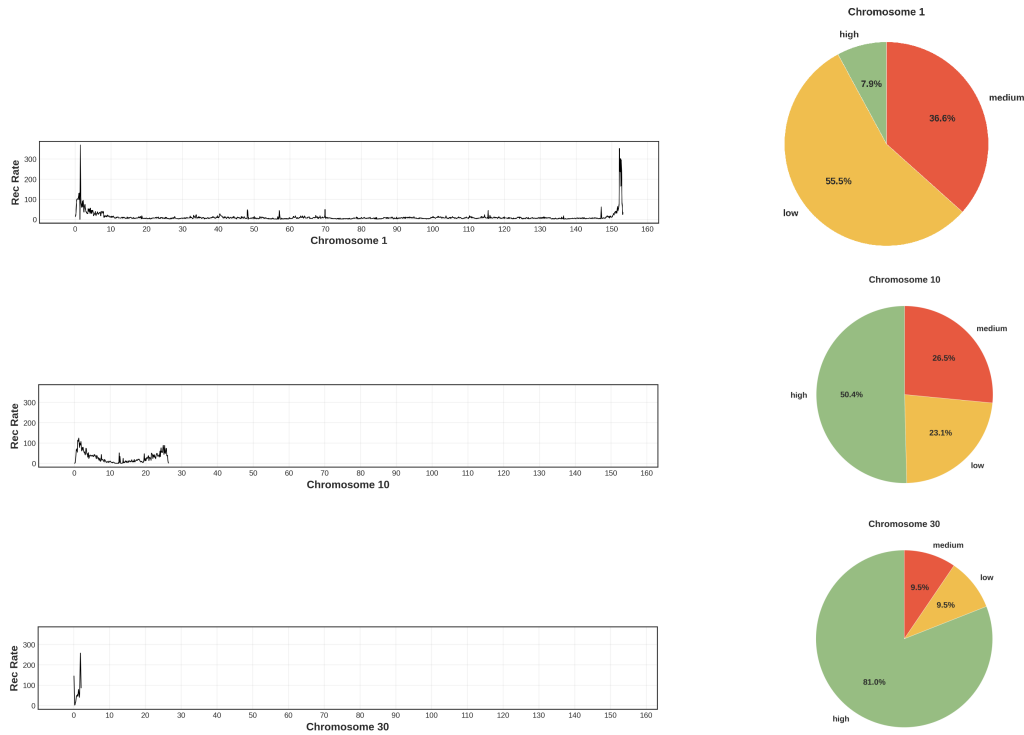


Figure 5. (Left): Recombination Rate ($\frac{cM}{Mb}$) across chromosomes 1, 10, and 30 in 100 kb windows. **(Right):** Recombination Rate binned into equal tertiles (*low*, *medium*, *high*).

Table 1. Genome-Wide and chromosome averages of European blackcaps genomic features

Chromosome	GC Content	Genes/Mb	Tajima D	SNP Density	Window π	cM/Mb
chr1	0.3920	11.0063	-1.2269	52.7326	0.0050	0.1330
chr2	0.3955	12.2712	-1.1908	55.0293	0.0054	0.1515
chr3	0.3988	13.2562	-1.2196	53.0504	0.0051	0.1323
chr4	0.3947	13.4057	-1.0386	58.2552	0.0061	0.1767
chr5	0.4036	15.53525	-1.0859	56.4039	0.0058	0.1842
chr6	0.4109	18.3237	-1.1007	55.0354	0.0057	0.2055
chr7	0.4151	15.9281	-1.1474	54.7267	0.0055	0.2541
chr8	0.4240	17.5278	-1.1360	56.6891	0.0057	0.2557
chr9	0.4275	20.4357	-1.1152	54.1897	0.0055	0.2565
chr10	0.4251	19.0993	-1.1387	53.1731	0.0055	0.3346
chr11	0.4273	18.7421	-0.9772	54.6924	0.0060	0.3228
chr12	0.4230	19.8661	-0.9134	50.9150	0.0056	0.2761
chr13	0.4400	21.7308	-1.1483	52.9053	0.0053	0.3238
chr14	0.4594	21.2564	-1.0125	57.3052	0.0061	0.3725
chr15	0.4616	29.8504	-0.9730	59.3889	0.0065	0.3987
chr16	0.4717	26.7848	-1.0122	57.2724	0.0061	0.4225
chr17	0.4746	28.6704	-0.9908	57.1325	0.0061	0.4154
chr18	0.4802	29.6759	-0.8893	63.1249	0.0071	0.4832
chr19	0.4891	28.1897	-0.9552	58.2750	0.0062	0.4800
chr20	0.4819	32.8829	-1.0120	56.3063	0.0061	0.5021
chr21	0.4521	24.7915	-0.9043	60.4145	0.0068	0.5539
chr22	0.4857	33.5450	-0.9848	56.8038	0.0062	0.6170
chr23	0.4991	30.3530	-1.0390	60.4349	0.0065	0.7416
chr24	0.5157	36.0876	-1.0019	61.3774	0.0067	0.7132
chr25	0.5230	43.5050	-1.0224	59.2732	0.0063	0.8062
chr26	0.5251	53.2288	-1.1627	53.9661	0.0054	0.7040
chr27	0.5197	61.1994	-1.1095	54.7559	0.0056	0.6322
chr28	0.5153	43.6125	-0.9401	63.2349	0.0069	0.9955
chr29	0.5610	65.4167	-1.2606	53.1565	0.0049	1.1710
chr30	0.5621	73.1818	-1.0367	51.8019	0.0054	0.7643
chr31	0.5306	59.5996	-1.5640	41.0650	0.0033	0.5380
chr32	0.4940	52.5460	-1.9144	19.0520	0.0011	0.2737
chr33	0.4957	136.1750	-0.8891	79.6533	0.0088	1.3076
Average	0.4540	33.8391	-1.0572	53.9049	0.0056	0.4778

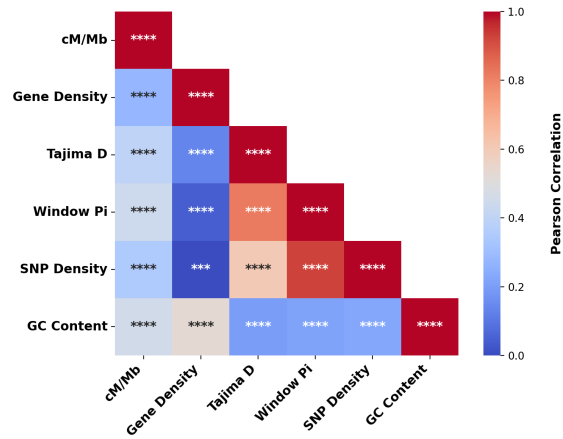


Figure 6. Heatmap showing the pearson correlation coefficient between several blackcap metapopulation features with corresponding p -values.

4.4 Wavelet transformation identifies meiotic recombination hotspots in blackcaps

To identify sharp peaks in the recombination rate, I applied a continuous wavelet transformation. Continuous wavelet transformations capture sharp, multi-scale differences in genomic signal intensity, making them suitable for detecting jumps in the recombination rate (González-Nuevo et al. 2006; G. R. Lee et al. 2019).

Utilizing wavelet transform to identify the ridges of the recombination rate signal with a peak prominence threshold of 90 and 30, I identified 712, and 6145 putative recombination hotspots (See Materials & Methods for details). The putative hotspots identified with a threshold of 90, are located mostly in the subtelomeric regions of chromosomes. Hotspots also appear along the chromosomal body, especially in the smaller chromosomes (Figure 8). The same trend is apparent in putative hotspots identified with a threshold of 30, although hotspots are also observed in the chromosomal body of the larger chromosomes, suggesting a broader distribution at lower detection thresholds (Figure 8, Supplementary Figures 9, 10).

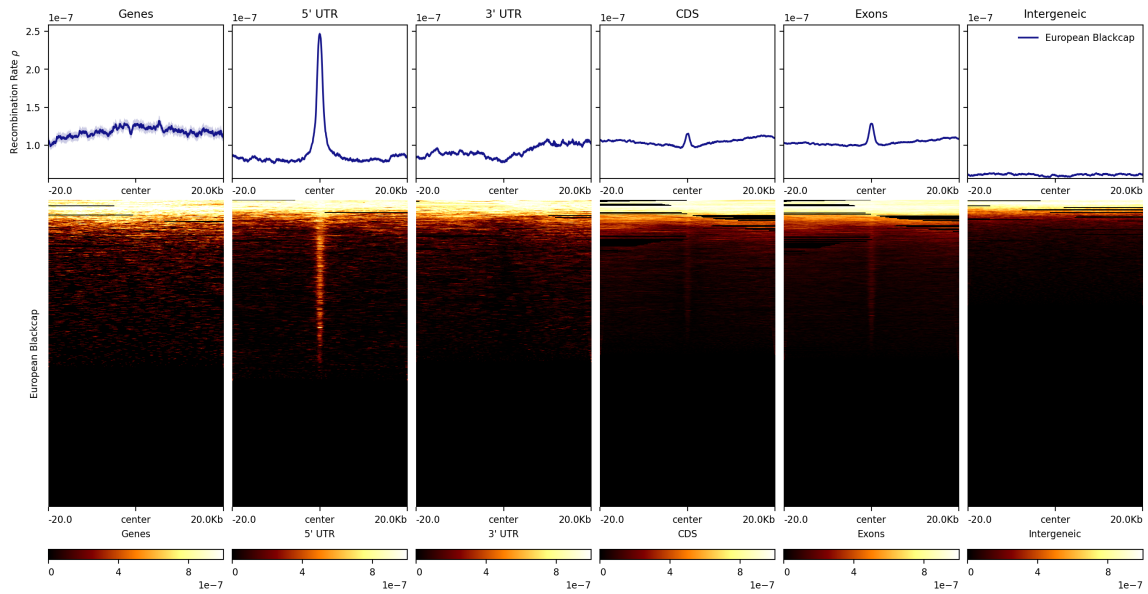


Figure 7. (Top) Average profile of the raw recombination rate ρ over centered functional genomic features (Genes, 5' UTRs, 3'UTRs, Exons, CDS and Intergenic regions) and **(bottom)** heatmap representing recombination rate ρ intensity per interval of functional genomic features computed on deepTools (Ramírez et al. 2016)

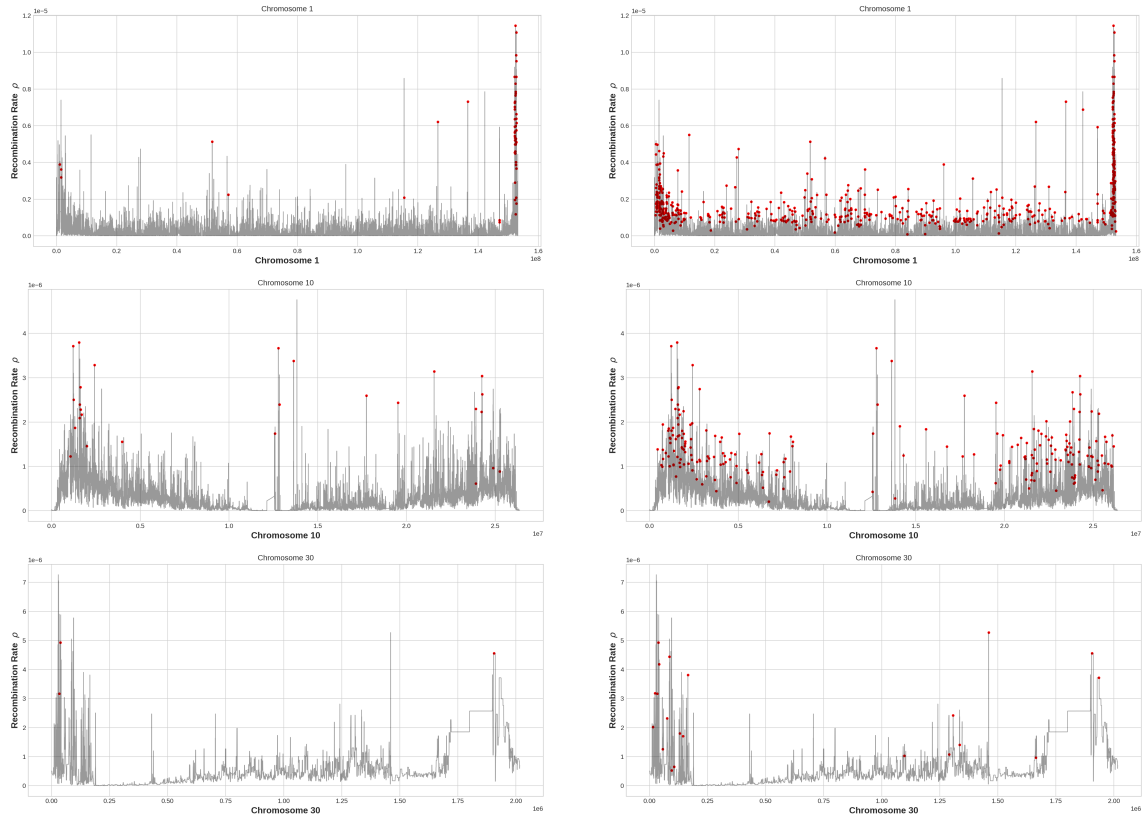


Figure 8. Gaussian smoothed ($\sigma = 5$) recombination rate on chromosomes 1, 10, 30 of black-caps with recombination hotspot peaks called by "mexican hat" wavelet transform (threshold 90) (**left**) and (threshold 30) (**right**).

To compare if functional genomic regions show the same pattern of recombination hotspots enrichment, as the recombination rate (Figure 7), I computed the Normalized pointwise mutual information between the identified hotspots and the functional genomic features (Table 2). The NPMI revealed that 5' UTRs, Genes, CDS, and Exons are slightly more enriched in recombination hotspots (called with peak prominence threshold 90), than what would be expected under a null distribution (Table 2, left). However hotspots called with a peak prominence threshold of 30, showed a higher NPMI, one order of magnitude higher for 5' UTRs, while the NPMI of hotspots on Genes, CDS and Exons remained relatively similar (Table 2, right). 3' UTRs and intergenic regions were found to be less enriched in recombination hotspots than expected by chance (Table 2).

Table 2. Normalized pointwise mutual information (NPMI) between Blackcap recombination hotspots called with wavelet transform using prominence threshold 90 (left) and 30 (right), measured against functional genomic features.

Threshold 90					Threshold 30				
Feature	NPMI	Exp.	LCI	UCI	Feature	NPMI	Exp.	LCI	UCI
3'UTR	-0.0369	0.0000	-0.0665	-0.0177	3'UTR	-0.0556	-0.0000	-0.0608	-0.0464
5'UTR	0.0396	0.0000	0.0106	0.0617	5'UTR	0.1694	0.0000	0.1670	0.1830
CDS	0.0460	0.0000	0.0176	0.0547	CDS	0.0602	0.0000	0.0530	0.0612
Exons	0.0225	0.0000	0.0064	0.0390	Exons	0.0618	0.0000	0.0573	0.0681
Genes	0.0157	0.0000	0.0050	0.0245	Genes	0.0247	0.0000	0.0208	0.0298
Intergenic	-0.0113	0.0000	-0.0181	-0.0082	Intergenic	-0.0160	0.0000	-0.0206	-0.0132

Exp. = expected NPMI under random overlaps. LCI/UCI = lower and upper 95% confidence interval bounds, respectively.

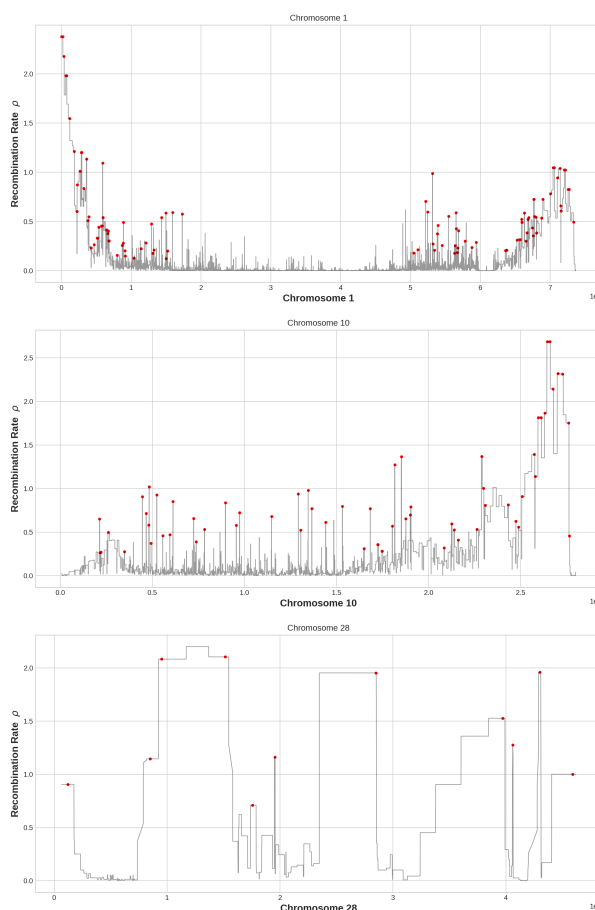


Figure 9. Gaussian smoothed ($\sigma = 5$) recombination rate on chromosomes 1, 10, 30 of zebra finches with recombination hotspot peaks called by "mexican hat" wavelet transform (threshold 30). Data from (Singhal et al. 2015).

5 Discussion

5.1 Recent population expansion and high diversity in the blackcap genome

When conducting a population genomics study, especially in non-model organisms such as *European blackcaps*, it is important to investigate general statistics such as Tajima's D , nucleotide diversity π and SNP density (Tajima 1989; Nei and Li 1979). These measures yield important information about the genome. Firstly, Nucleotide diversity π measures the average pairwise differences per site between chromosomes, which reflects the amount of variation retained within a genomic region (Nielsen and M. Slatkin 2013; Nei and Li 1979). High π is an indicator of effective breaks in the linkage between sites, which is a result of high recombination. Conversely, low π is expected in low recombining regions where background selection or hitchhiking are more prominent. The blackcap population has a genome wide nucleotide π of 0.0056, which is 0.44 percentage points higher than in humans ($\pi = 0.0011$) (Table 1) (Li and Sadler 1991). Furthermore, the π values were higher in the microchromosomes, with chromosome 33 having the highest value of 0.0088 (Table 1). These high π values imply high recombination activity in the genomes of blackcaps.

Secondly, Tajima's D , another statistic in population genomics, measures the difference between the nucleotide π and the number of segregating sites θ_w , mirroring the shape of the allele frequency spectrum (Nielsen and M. Slatkin 2013; Tajima 1989). A negative D is an indicator of excess low frequency alleles, that arise from recent selective sweeps, reduced recombination or recent population expansions (Nielsen and M. Slatkin 2013; Tajima 1989). Values near zero reflect neutral equilibrium while positive values result from balancing selection. The blackcap genome has an average Tajima's D of -1.0572, which remains relatively stable across chromosomes (Table 1). Local negative Tajima's D values could indicate selective sweeps or reduced recombination, however since the Tajima's D doesn't show extreme drops across the chromosomes and retains a negative value it indicates a recent population expansion event or low frequency alleles that are continuously generated and removed via background selection and/or recombination events (Figure 1, Supplementary Figure 1).

Thirdly, the SNP density represents the number of polymorphic sites per genomic window (Nielsen and M. Slatkin 2013). Regions with high SNP Density reflect high recombination or weaker purifying selection, and vice versa. The blackcap genome has $\approx 54 \frac{\text{Variants}}{\text{kb}}$; the microchromosomes having a higher SNP density than macrochromosomes (Table 1, Figure 2). As

high SNP density is indicative of standing variation.

Lastly, incorporating these three measurements together one can suggest that the blackcaps are a large, high recombining population that maintain their high genetic diversity with a recent demographic expansion.

5.2 Strong correlation between gc content and recombination rate in blackcaps

The GC content has been repeatedly shown to have a strong correlation with the recombination rate in organisms from yeast to humans (Kiktev et al. 2018; Chen et al. 2006; Fullerton, Bernardo Carvalho, and Clark 2001). To check if this is also the case in blackcaps I computed the gc content in 100 kb windows and computed the pearson correlation coefficient (Figure 6). The resulting value of $r \approx 0.55$ is consistent with what has been reported before in humans, mice and yeasts. Mainly strong correlation between gc content and recombination rate (Kiktev et al. 2018; Chen et al. 2006; Fullerton, Bernardo Carvalho, and Clark 2001). There are several possibilities that could explain this correlation.

Firstly, during the repair of DNA double stranded breaks needed to form recombination hotspots a bias towards preferential DSB repair with GC over AT nucleotides could arise, which is known as GC Biased Gene Conversion (Bolívar et al. 2019; Duret and Galtier 2009). Secondly, blackcaps could have a mutation bias towards GC nucleotides. Such events where organisms have preferential dinucleotides to which their genome mutates over time have been observed in both prokaryotes (Hershberg and Petrov 2010). Thirdly, the high GC content on the blackcap genome could be due to selection for stable DNA. GC pairs have 3 hydrogen bonds as opposed to 2 between AT pairs, thus rendering them more stable towards excessive double stranded breaks as birds do not have the *prdm9* gene, which also plays a role in controlling excessive DNA breaks during recombination (Paiano et al. 2020; Mustonen and Lässig 2005). Lastly, the high GC Content could be attributed to high gene density and functional elements (Bird 1986), which is supported by a correlation coefficient of $r \approx 0.7$ (Figure 6)

5.3 Blackcap 5' UTRs are enriched in recombination hotspots

In organisms carrying *prdm9*, recombination is directed away from functional genomic elements, by the trimethylation of histones 3 lysine 4 (H3K4me3) and histone 3 lysine 36 (H3K36me3)

(Cavassim et al. 2022; Parvanov et al. 2017; Lange et al. 2016; Baker et al. 2015; Berg et al. 2010). However birds lack the *prdm9* gene completely (Cavassim et al. 2022). To investigate if the absence of *prdm9* leads to enriched recombination rate in functional genomic features I computed the enrichment and average profile of the recombination rate over genes, UTR's, CDS, exons and intergenic regions. The recombination rate had a sharp prominent peak in the center of 5' UTRs ($\rho \approx 2.5^{10-7}$), followed by small peaks on cds and exons while showing weak or no signal intergenic regions and the center of genes (Figure 7).

An important observation is that the recombination rate is elevated in subtelomeric regions, with some peaks along the chromosomal axis, characteristics which have been previously described in zebra finches and long tailed finches, pointing towards possible evolutionary conservation of recombination hotspots (Singhal et al. 2015). However this hypothesis could not be tested due to the absence of files needed to map the genomic intervals of finches to the genome blackcaps caused by lack of funding in the team of UCSC Genome Browser needed to generate such files.

Recombination maps, computed by *pyrho*, convey information about all historic recombination activity across the genome, because the calculation incorporates demographic history information to infer recombination rates (Spence and Song 2019). As recombination activity is turned over with time, the most recent recombination hotspots are expected to have higher recombination rates than older hotspots (Úbeda and Wilkins 2011). So one needs to identify large jumps in recombination rate values over a local background. Previously recombination hotspots were called from recombination maps inferred through LD methods with a sliding window approach (Wooldridge and Dumont 2023). However, the sliding window approach lacks multi scale resolution and risks flattening peaks as it computes an average over a large window, losing fine scale resolution. To circumvent these limitations I applied the continuous wavelet transformation (González-Nuevo et al. 2006).

The continuous wavelet transformation consists of a "mother" wavelet function that is slid across a signal over multiple scales and computes how well a signal fits the wavelet at that position and scale. After computing the wavelet coefficient, I retained the coefficients that persisted over at least 30% of the scales. Next I applied another level of thresholding by computing the per scale Z-Score and selecting only those coefficients with $Z > 2.0$. Finally, the coefficients repre-

senting the same peak over several scales are collapsed into a single coefficient, that is used to call identify peaks based on its prominence over neighbouring coefficients.

I was able to identify 712 putative hotspots when using a peak prominence threshold equal to the 90th percentile of recombination rate values and 6145 when using a threshold of 30. When computing the normalized pointwise mutual information, it was clearly seen that hotspots followed the trend of having the highest enrichment in 5' UTRs as opposed to other genomic features, although this signal was more prominent in the hotspots identified with a threshold of 30 (Table 2). This jump in NPMI in the lower threshold of 30, which is observed between hotspots and 5' UTRs could be an artifact of the mathematics behind normalized mutual pointwise information, which when comparing the quantifying the enrichment of only a few intervals (712 windows) within a larger quantity of windows such as 5' UTRs (42816 windows) might underestimate the true value

To test if this methodology is also applicable to other datasets which were not generated through pyrho, I utilized the recombination rate maps from Singhal et al. 2015 that were generated through LDhot. In this case the wavelet coefficients still identified recombination hotspots which were mostly localized in the subtelomeric regions with occasional ones along the chromosomal axis (Figure 9)

5.4 Conclusions

This thesis sought to extend the analysis of the meiotic recombination landscape of blackcap populations from 30 individuals (Bascón-Cardozo et al. 2024) to 100 as well as establish a novel, accurate methodology for calling present recombination hotspots from historic recombination rate data. Using population genome statistics such as Tajima's D , nucleotide π and snp density; Blackcap GFF files (containing location of genes, UTRs, exons and codons), computing the GC content and gene density as well the recombination rate with pyrho, I was able to uncover that the blackcap genome although it shows recent population expansions still has relatively high diversity, high gc content and high recombination activity (Table 1). Furthermore these features showed extreme values at subtelomeric regions (Figures 1 - 5, Supplementary Figures 1-6), with the recombination rate, gene density, gc content, π and gene density increasing in the microchromosomes. Recombination rate achieved the highest average enrichment over 5'

UTRs ($\rho = 2.5 \times 10^{-7}$), with small peaks in codons and exons ($\rho \approx 1.2 \times 10^{-7}$) while no recombination rate was found in intragenic regions as it has been observed in species carrying *prdm9* (Wooldridge and Dumont 2023; Lange et al. 2016; Baker et al. 2015). Furthermore the presence of strong recombination rate signal in the subtelomeric regions mirrored findings in zebra finches and long tailed finches (Singhal et al. 2015). Finally wavelet transformation provides a novel, more accurate and computationally efficient methodology to identify hotspots from recombination rate maps, as they are able to analyze the signal at a multi-scale level thus yielding fine scale resolution and robust noise handling (Figure 8) as well as being persistent in datasets generated by LD methods (Figure 9)(Singhal et al. 2015)).

6 References

- AbuAlia, Khawla F N et al. (Mar. 2024). “Natural Variation in the Zinc-Finger-Encoding Exon of Prdm9 Affects Hybrid Sterility Phenotypes in Mice”. In: *Genetics* 226.3, iyae004. ISSN: 1943-2631. DOI: [10.1093/genetics/iyae004](https://doi.org/10.1093/genetics/iyae004). URL: <https://doi.org/10.1093/genetics/iyae004> (visited on 05/14/2024).
- Auton, Adam and Gil McVean (Aug. 1, 2007). “Recombination rate estimation in the presence of hotspots”. In: *Genome Research* 17.8. Company: Cold Spring Harbor Laboratory Press Distributor: Cold Spring Harbor Laboratory Press Institution: Cold Spring Harbor Laboratory Press Label: Cold Spring Harbor Laboratory Press Publisher: Cold Spring Harbor Lab, pp. 1219–1227. ISSN: 1088-9051, 1549-5469. DOI: [10.1101/gr.6386707](https://doi.org/10.1101/gr.6386707). URL: <http://genome.cshlp.org/content/17/8/1219> (visited on 09/18/2025).
- Baker, Christopher L. et al. (Jan. 2015). “PRDM9 Drives Evolutionary Erosion of Hotspots in *Mus musculus* through Haplotype-Specific Initiation of Meiotic Recombination”. en. In: *PLOS Genetics* 11.1. Publisher: Public Library of Science, e1004916. ISSN: 1553-7404. DOI: [10.1371/journal.pgen.1004916](https://doi.org/10.1371/journal.pgen.1004916). URL: <https://journals.plos.org/plosgenetics/article?id=10.1371/journal.pgen.1004916> (visited on 09/06/2024).
- Bascón-Cardozo, Karen et al. (Jan. 2024). “Fine-Scale Map Reveals Highly Variable Recombination Rates Associated with Genomic Features in the Eurasian Blackcap”. In: *Genome Biology and Evolution* 16.1, evad233. ISSN: 1759-6653. DOI: [10.1093/gbe/evad233](https://doi.org/10.1093/gbe/evad233). URL: <https://doi.org/10.1093/gbe/evad233> (visited on 04/06/2025).

-
- Baudat, Frédéric, Yukiko Imai, and Bernard de Massy (Nov. 2013). “Meiotic recombination in mammals: localization and regulation”. en. In: *Nature Reviews Genetics* 14.11. Publisher: Nature Publishing Group, pp. 794–806. ISSN: 1471-0064. DOI: [10.1038/nrg3573](https://doi.org/10.1038/nrg3573). URL: <https://www.nature.com/articles/nrg3573> (visited on 11/10/2024).
- Baumdicker, Franz et al. (Mar. 1, 2022). “Efficient ancestry and mutation simulation with msprime 1.0”. In: *Genetics* 220.3, iyab229. ISSN: 1943-2631. DOI: [10.1093/genetics/iyab229](https://doi.org/10.1093/genetics/iyab229). URL: <https://doi.org/10.1093/genetics/iyab229> (visited on 09/19/2025).
- Berg, Ingrid L. et al. (Oct. 2010). “PRDM9 variation strongly influences recombination hot-spot activity and meiotic instability in humans”. en. In: *Nature Genetics* 42.10. Publisher: Nature Publishing Group, pp. 859–863. ISSN: 1546-1718. DOI: [10.1038/ng.658](https://doi.org/10.1038/ng.658). URL: <https://www.nature.com/articles/ng.658> (visited on 07/15/2025).
- Bird, Adrian P. (1986). “CpG-rich islands and the function of DNA methylation”. In: *Nature* 321.6067, pp. 209–213. ISSN: 1476-4687. DOI: [10.1038/321209a0](https://doi.org/10.1038/321209a0). URL: <https://doi.org/10.1038/321209a0>.
- Bolívar, Paulina et al. (2019). “GC-biased gene conversion conceals the prediction of the nearly neutral theory in avian genomes”. In: *Genome Biology* 20.1, p. 5. ISSN: 1474-760X. DOI: [10.1186/s13059-018-1613-z](https://doi.org/10.1186/s13059-018-1613-z). URL: <https://doi.org/10.1186/s13059-018-1613-z>.
- Börner, G. Valentin, Nancy Kleckner, and Neil Hunter (Apr. 2004). “Crossover/Noncrossover Differentiation, Synaptonemal Complex Formation, and Regulatory Surveillance at the Leptotene/ Zygotene Transition of Meiosis”. In: *Cell* 117.1, pp. 29–45. ISSN: 0092-8674. DOI: [10.1016/S0092-8674\(04\)00292-2](https://doi.org/10.1016/S0092-8674(04)00292-2). URL: <https://www.sciencedirect.com/science/article/pii/S0092867404002922> (visited on 05/24/2024).
- Bours, Andrea et al. (Sept. 2023). “The blackcap (*Sylvia atricapilla*) genome reveals a recent accumulation of LTR retrotransposons”. en. In: *Scientific Reports* 13.1. Publisher: Nature Publishing Group, p. 16471. ISSN: 2045-2322. DOI: [10.1038/s41598-023-43090-1](https://doi.org/10.1038/s41598-023-43090-1). URL: <https://www.nature.com/articles/s41598-023-43090-1> (visited on 03/04/2025).
- Cavassim, Maria Izabel A. et al. (Mar. 2022). “PRDM9 losses in vertebrates are coupled to those of paralogs ZCWPW1 and ZCWPW2”. In: *Proceedings of the National Academy of Sciences* 119.9. Publisher: Proceedings of the National Academy of Sciences, e2114401119. DOI: [10.1073/pnas.2114401119](https://doi.org/10.1073/pnas.2114401119). URL: <https://www.pnas.org/doi/full/10.1073/pnas.2114401119> (visited on 11/04/2024).

-
- Chan, Andrew H., Paul A. Jenkins, and Yun S. Song (Dec. 2012). “Genome-Wide Fine-Scale Recombination Rate Variation in *Drosophila melanogaster*”. In: *PLOS Genetics* 8.12, pp. 1–28. DOI: [10.1371/journal.pgen.1003090](https://doi.org/10.1371/journal.pgen.1003090). URL: <https://doi.org/10.1371/journal.pgen.1003090>.
- Chen, Jin-Feng et al. (2006). “Significant positive correlation between the recombination rate and GC content in the human pseudoautosomal region”. In: *Genome* 49.5. PMID: 16767166, pp. 413–419. DOI: [10.1139/g05-124](https://doi.org/10.1139/g05-124). eprint: <https://doi.org/10.1139/g05-124>. URL: <https://doi.org/10.1139/g05-124>.
- Danecek, Petr et al. (June 2011). “The variant call format and VCFtools”. In: *Bioinformatics* 27.15, pp. 2156–2158. ISSN: 1367-4803. DOI: [10.1093/bioinformatics/btr330](https://doi.org/10.1093/bioinformatics/btr330). eprint: https://academic.oup.com/bioinformatics/article-pdf/27/15/2156/48863483/bioinformatics_27_15_2156.pdf. URL: <https://doi.org/10.1093/bioinformatics/btr330>.
- Duret, Laurent and Nicolas Galtier (2009). “Biased gene conversion and the evolution of mammalian genomic landscapes”. In: *Annual Review of Genomics and Human Genetics* 10, pp. 285–311. ISSN: 1545-293X. DOI: [10.1146/annurev-genom-082908-150001](https://doi.org/10.1146/annurev-genom-082908-150001).
- Fullerton, Stephanie M., Antonio Bernardo Carvalho, and Andrew G. Clark (June 2001). “Local Rates of Recombination Are Positively Correlated with GC Content in the Human Genome”. In: *Molecular Biology and Evolution* 18.6, pp. 1139–1142. ISSN: 0737-4038. DOI: [10.1093/oxfordjournals.molbev.a003886](https://doi.org/10.1093/oxfordjournals.molbev.a003886). eprint: <https://academic.oup.com/mbe/article-pdf/18/6/1139/23448607/i0737-4038-018-06-1139.pdf>. URL: <https://doi.org/10.1093/oxfordjournals.molbev.a003886>.
- González-Nuevo, J. et al. (July 2006). “The Mexican hat wavelet family: application to point-source detection in cosmic microwave background maps”. In: *Monthly Notices of the Royal Astronomical Society* 369.4, pp. 1603–1610. ISSN: 0035-8711. DOI: [10.1111/j.1365-2966.2006.10442.x](https://doi.org/10.1111/j.1365-2966.2006.10442.x). eprint: <https://academic.oup.com/mnras/article-pdf/369/4/1603/3797826/mnras0369-1603.pdf>. URL: <https://doi.org/10.1111/j.1365-2966.2006.10442.x>.
- Handel, Mary Ann and John C. Schimenti (Feb. 2010). “Genetics of mammalian meiosis: regulation, dynamics and impact on fertility”. en. In: *Nature Reviews Genetics* 11.2. Publisher: Nature Publishing Group, pp. 124–136. ISSN: 1471-0064. DOI: [10.1038/nrg2723](https://doi.org/10.1038/nrg2723). URL: <https://www.nature.com/articles/nrg2723> (visited on 07/15/2025).

-
- Hershberg, Ruth and Dmitri A. Petrov (2010). “Evidence that mutation is universally biased towards AT in bacteria”. In: *PLoS Genetics* 6.9, e1001115. ISSN: 1553-7390. DOI: [10.1371/journal.pgen.1001115](https://doi.org/10.1371/journal.pgen.1001115).
- Hinch, Anjali Gupta et al. (Aug. 2020). “The Configuration of RPA, RAD51, and DMC1 Binding in Meiosis Reveals the Nature of Critical Recombination Intermediates”. English. In: *Molecular Cell* 79.4. Number: 4 Publisher: Elsevier, 689–701.e10. ISSN: 1097-2765. DOI: [10.1016/j.molcel.2020.06.015](https://doi.org/10.1016/j.molcel.2020.06.015). URL: [https://www.cell.com/molecular-cell/abstract/S1097-2765\(20\)30400-7](https://www.cell.com/molecular-cell/abstract/S1097-2765(20)30400-7) (visited on 07/28/2022).
- Hinrichs, A. S. et al. (Jan. 2006). “The UCSC Genome Browser Database: update 2006”. In: *Nucleic Acids Research* 34.suppl_1, pp. D590–D598. ISSN: 0305-1048. DOI: [10.1093/nar/gkj144](https://doi.org/10.1093/nar/gkj144). eprint: https://academic.oup.com/nar/article-pdf/34/suppl_1/D590/3926288/gkj144.pdf. URL: <https://doi.org/10.1093/nar/gkj144>.
- Hunter, Neil (Jan. 2015). “Meiotic Recombination: The Essence of Heredity”. en. In: *Cold Spring Harbor Perspectives in Biology* 7.12. Company: Cold Spring Harbor Laboratory Press Distributor: Cold Spring Harbor Laboratory Press Institution: Cold Spring Harbor Laboratory Press Label: Cold Spring Harbor Laboratory Press Publisher: Cold Spring Harbor Lab, a016618. ISSN: , 1943-0264. DOI: [10.1101/cshperspect.a016618](https://doi.org/10.1101/cshperspect.a016618). URL: <http://cshperspectives.cshlp.org/content/7/12/a016618> (visited on 05/21/2024).
- Imai, Yukiko et al. (Dec. 2017). “The PRDM9 KRAB domain is required for meiosis and involved in protein interactions”. en. In: *Chromosoma* 126.6, pp. 681–695. ISSN: 1432-0886. DOI: [10.1007/s00412-017-0631-z](https://doi.org/10.1007/s00412-017-0631-z). URL: <https://doi.org/10.1007/s00412-017-0631-z> (visited on 05/23/2024).
- Ishigohoka, Jun and Miriam Liedvogel (Jan. 2025). “High-recombining genomic regions affect demography inference based on ancestral recombination graphs”. In: *Genetics*, iyaf004. ISSN: 1943-2631. DOI: [10.1093/genetics/iyaf004](https://doi.org/10.1093/genetics/iyaf004). URL: <https://doi.org/10.1093/genetics/iyaf004> (visited on 03/03/2025).
- Johnston, Susan E (July 2024). “Understanding the Genetic Basis of Variation in Meiotic Recombination: Past, Present, and Future”. In: *Molecular Biology and Evolution* 41.7, msae112. ISSN: 1537-1719. DOI: [10.1093/molbev/msae112](https://doi.org/10.1093/molbev/msae112). eprint: <https://academic.oup.com/mbe/article-pdf/41/7/msae112/58685648/msae112.pdf>. URL: <https://doi.org/10.1093/molbev/msae112>.

-
- Joseph, Julien et al. (June 2024). “High prevalence of PRDM9-independent recombination hotspots in placental mammals”. en. In: *Proceedings of the National Academy of Sciences* 121.23, e2401973121. ISSN: 0027-8424, 1091-6490. DOI: [10 . 1073 / pnas . 2401973121](https://doi.org/10.1073/pnas.2401973121). URL: <https://pnas.org/doi/10.1073/pnas.2401973121> (visited on 01/11/2025).
- Kiktev, Denis A. et al. (2018). “GC content elevates mutation and recombination rates in the yeast *Saccharomyces cerevisiae*”. In: *Proceedings of the National Academy of Sciences* 115.30, E7109–E7118. DOI: [10 . 1073 / pnas . 1807334115](https://doi.org/10.1073/pnas.1807334115). eprint: <https://www.pnas.org/doi/pdf/10.1073/pnas.1807334115>. URL: <https://www.pnas.org/doi/abs/10.1073/pnas.1807334115>.
- Lam, Kwan-Wood Gabriel et al. (Aug. 2019). “Cell-type-specific genomics reveals histone modification dynamics in mammalian meiosis”. en. In: *Nature Communications* 10.1. Number: 1 Publisher: Nature Publishing Group, p. 3821. ISSN: 2041-1723. DOI: [10 . 1038 / s41467 - 019 - 11820 - 7](https://doi.org/10.1038/s41467-019-11820-7). URL: <https://www.nature.com/articles/s41467-019-11820-7> (visited on 01/16/2024).
- Lange, Julian et al. (Oct. 2016). “The Landscape of Mouse Meiotic Double-Strand Break Formation, Processing, and Repair”. English. In: *Cell* 167.3. Publisher: Elsevier, 695–708.e16. ISSN: 0092-8674, 1097-4172. DOI: [10 . 1016 / j . cell . 2016 . 09 . 035](https://doi.org/10.1016/j.cell.2016.09.035). URL: [https://www.cell.com/cell/abstract/S0092-8674\(16\)31317-4](https://www.cell.com/cell/abstract/S0092-8674(16)31317-4) (visited on 01/16/2024).
- Lee, Gregory R. et al. (2019). “PyWavelets: A Python package for wavelet analysis”. In: *Journal of Open Source Software* 4.36, p. 1237. DOI: [10 . 21105 / joss . 01237](https://doi.org/10.21105/joss.01237). URL: <https://doi.org/10.21105/joss.01237>.
- Lee, Junghee et al. (July 2019). “EWSR1, a multifunctional protein, regulates cellular function and aging via genetic and epigenetic pathways”. In: *Biochimica et Biophysica Acta (BBA) - Molecular Basis of Disease*. Genetic and epigenetic regulation of aging and longevity 1865.7, pp. 1938–1945. ISSN: 0925-4439. DOI: [10 . 1016 / j . bbadis . 2018 . 10 . 042](https://doi.org/10.1016/j.bbadis.2018.10.042). URL: <https://www.sciencedirect.com/science/article/pii/S0925443918304794> (visited on 06/30/2024).
- Li, W H and L A Sadler (Oct. 1991). “Low nucleotide diversity in man.” In: *Genetics* 129.2, pp. 513–523. ISSN: 1943-2631. DOI: [10 . 1093 / genetics / 129 . 2 . 513](https://doi.org/10.1093/genetics/129.2.513). eprint: <https://academic.oup.com/genetics/article-pdf/129/2/513/34556637/genetics0513.pdf>. URL: <https://doi.org/10.1093/genetics/129.2.513>.

-
- Ma, Tao et al. (Jan. 2023). “Cobind: quantitative analysis of the genomic overlaps”. In: *Bioinformatics Advances* 3.1, vbad104. ISSN: 2635-0041. DOI: [10.1093/bioadv/vbad104](https://doi.org/10.1093/bioadv/vbad104). URL: <https://doi.org/10.1093/bioadv/vbad104> (visited on 09/23/2024).
- McVean, Gil, Philip Awadalla, and Paul Fearnhead (Mar. 2002). “A Coalescent-Based Method for Detecting and Estimating Recombination From Gene Sequences”. In: *Genetics* 160.3, pp. 1231–1241. ISSN: 1943-2631. DOI: [10.1093/genetics/160.3.1231](https://doi.org/10.1093/genetics/160.3.1231). eprint: <https://academic.oup.com/genetics/article-pdf/160/3/1231/42036591/genetics1231.pdf>. URL: <https://doi.org/10.1093/genetics/160.3.1231>.
- McVean, Gilean A T (Oct. 1, 2002). “A Genealogical Interpretation of Linkage Disequilibrium”. In: *Genetics* 162.2, pp. 987–991. ISSN: 1943-2631. DOI: [10.1093/genetics/162.2.987](https://doi.org/10.1093/genetics/162.2.987). URL: <https://academic.oup.com/genetics/article/162/2/987/6049967> (visited on 09/19/2025).
- Mustonen, Ville and Michael Lässig (2005). “Evolutionary population genetics of promoters: Predicting binding sites and functional phylogenies”. In: *Proceedings of the National Academy of Sciences* 102.44, pp. 15936–15941. DOI: [10.1073/pnas.0505537102](https://doi.org/10.1073/pnas.0505537102). eprint: <https://www.pnas.org/doi/pdf/10.1073/pnas.0505537102>. URL: <https://www.pnas.org/doi/abs/10.1073/pnas.0505537102>.
- Nei, M and W H Li (1979). “Mathematical model for studying genetic variation in terms of restriction endonucleases.” In: *Proceedings of the National Academy of Sciences* 76.10, pp. 5269–5273. DOI: [10.1073/pnas.76.10.5269](https://doi.org/10.1073/pnas.76.10.5269). eprint: <https://www.pnas.org/doi/pdf/10.1073/pnas.76.10.5269>. URL: <https://www.pnas.org/doi/abs/10.1073/pnas.76.10.5269>.
- Nielsen, R. and M. Slatkin (2013). *An Introduction to Population Genetics: Theory and Applications*. Sinauer. ISBN: 9781605351537. URL: <https://books.google.de/books?id=Iy08kgEACAAJ>.
- Paiano, Jacob et al. (Feb. 2020). “ATM and PRDM9 regulate SPO11-bound recombination intermediates during meiosis”. en. In: *Nature Communications* 11.1. Number: 1 Publisher: Nature Publishing Group, p. 857. ISSN: 2041-1723. DOI: [10.1038/s41467-020-14654-w](https://doi.org/10.1038/s41467-020-14654-w). URL: <https://www.nature.com/articles/s41467-020-14654-w> (visited on 01/16/2024).
- Parvanov, Emil D. et al. (June 2016). *PRDM9 forms a multiprotein complex tethering recombination hotspots to the chromosomal axis*. en. DOI: [10.1101/056713](https://doi.org/10.1101/056713). URL: <http://biorxiv.org/lookup/doi/10.1101/056713> (visited on 04/19/2024).

-
- Parvanov, Emil D. et al. (Feb. 2017). “PRDM9 interactions with other proteins provide a link between recombination hotspots and the chromosomal axis in meiosis”. In: *Molecular Biology of the Cell* 28.3. Publisher: American Society for Cell Biology (mboc), pp. 488–499. ISSN: 1059-1524. DOI: [10.1091/mbc.e16-09-0686](https://doi.org/10.1091/mbc.e16-09-0686). URL: <https://www.molbiolcell.org/doi/full/10.1091/mbc.e16-09-0686> (visited on 07/24/2024).
- Perez, Gerardo et al. (Oct. 2024). “The UCSC Genome Browser database: 2025 update”. In: *Nucleic Acids Research* 53.D1, pp. D1243–D1249. ISSN: 1362-4962. DOI: [10.1093/nar/gkae974](https://doi.org/10.1093/nar/gkae974). eprint: <https://academic.oup.com/nar/article-pdf/53/D1/D1243/60120664/gkae974.pdf>. URL: <https://doi.org/10.1093/nar/gkae974>.
- Pigozzi, María I. (Dec. 2016). “The Chromosomes of Birds during Meiosis”. In: *Cytogenetic and Genome Research* 150.2, pp. 128–138. ISSN: 1424-8581. DOI: [10.1159/000453541](https://doi.org/10.1159/000453541). URL: <https://doi.org/10.1159/000453541> (visited on 11/04/2024).
- Powers, Natalie R. et al. (June 2016). “The Meiotic Recombination Activator PRDM9 Trimethylates Both H3K36 and H3K4 at Recombination Hotspots In Vivo”. en. In: *PLOS Genetics* 12.6. Publisher: Public Library of Science, e1006146. ISSN: 1553-7404. DOI: [10.1371/journal.pgen.1006146](https://doi.org/10.1371/journal.pgen.1006146). URL: <https://journals.plos.org/plosgenetics/article?id=10.1371/journal.pgen.1006146> (visited on 01/16/2024).
- Ramírez, Fidel et al. (July 2016). “deepTools2: a next generation web server for deep-sequencing data analysis”. en. In: *Nucleic Acids Research* 44.W1, W160–W165. ISSN: 0305-1048, 1362-4962. DOI: [10.1093/nar/gkw257](https://doi.org/10.1093/nar/gkw257). URL: <https://academic.oup.com/nar/article-lookup/doi/10.1093/nar/gkw257> (visited on 12/01/2024).
- Sayers, Eric W et al. (Nov. 2024). “Database resources of the National Center for Biotechnology Information in 2025”. In: *Nucleic Acids Research* 53.D1, pp. D20–D29. ISSN: 1362-4962. DOI: [10.1093/nar/gkae979](https://doi.org/10.1093/nar/gkae979). eprint: <https://academic.oup.com/nar/article-pdf/53/D1/D20/60586411/gkae979.pdf>. URL: <https://doi.org/10.1093/nar/gkae979>.
- Singh, Abhishek, Aparna Rawat, and Nikhila Raghuthaman (2022). “Mexican Hat Wavelet Transform and Its Applications”. In: *Methods of Mathematical Modelling and Computation for Complex Systems*. Ed. by Jagdev Singh et al. Cham: Springer International Publishing, pp. 299–317. ISBN: 978-3-030-77169-0. DOI: [10.1007/978-3-030-77169-0_12](https://doi.org/10.1007/978-3-030-77169-0_12). URL: https://doi.org/10.1007/978-3-030-77169-0_12.
- Singhal, Sonal et al. (2015). “Stable recombination hotspots in birds”. In: *Science* 350.6263, pp. 928–932. DOI: [10.1126/science.aad0843](https://doi.org/10.1126/science.aad0843). eprint: <https://www.science.org/>

-
- [doi/pdf/10.1126/science.aad0843](https://doi.org/10.1126/science.aad0843). URL: <https://www.science.org/doi/abs/10.1126/science.aad0843>.
- Slatkin, Montgomery (June 2008). “Linkage disequilibrium — understanding the evolutionary past and mapping the medical future”. In: *Nature Reviews Genetics* 9.6, pp. 477–485. ISSN: 1471-0056, 1471-0064. DOI: [10.1038/nrg2361](https://doi.org/10.1038/nrg2361). URL: <https://www.nature.com/articles/nrg2361> (visited on 09/18/2025).
- Spence, Jeffrey P. and Yun S. Song (2019). “Inference and analysis of population-specific fine-scale recombination maps across 26 diverse human populations”. In: *Science Advances* 5.10, eaaw9206. DOI: [10.1126/sciadv.aaw9206](https://doi.org/10.1126/sciadv.aaw9206). eprint: <https://www.science.org/doi/pdf/10.1126/sciadv.aaw9206>. URL: <https://www.science.org/doi/abs/10.1126/sciadv.aaw9206>.
- Tajima, F (Nov. 1989). “Statistical method for testing the neutral mutation hypothesis by DNA polymorphism.” In: *Genetics* 123.3, pp. 585–595. ISSN: 1943-2631. DOI: [10.1093/genetics/123.3.585](https://doi.org/10.1093/genetics/123.3.585). eprint: <https://academic.oup.com/genetics/article-pdf/123/3/585/34466400/genetics0585.pdf>. URL: <https://doi.org/10.1093/genetics/123.3.585>.
- Tian, Hui, Timothy Billings, and Petko M. Petkov (Jan. 2021). “EWSR1 affects PRDM9-dependent histone 3 methylation and provides a link between recombination hotspots and the chromosome axis protein REC8”. In: *Molecular Biology of the Cell* 32.1. Publisher: American Society for Cell Biology (mboc), pp. 1–14. ISSN: 1059-1524. DOI: [10.1091/mbc.E20-09-0604](https://doi.org/10.1091/mbc.E20-09-0604). URL: <https://www.molbiolcell.org/doi/10.1091/mbc.E20-09-0604> (visited on 03/19/2024).
- Úbeda, F. and J. F. Wilkins (Mar. 2011). “The Red Queen theory of recombination hotspots”. In: *Journal of Evolutionary Biology* 24.3, pp. 541–553. ISSN: 1010-061X. DOI: [10.1111/j.1420-9101.2010.02187.x](https://doi.org/10.1111/j.1420-9101.2010.02187.x). eprint: <https://academic.oup.com/jeb/article-pdf/24/3/541/54172857/jevbio0541.pdf>. URL: <https://doi.org/10.1111/j.1420-9101.2010.02187.x>.
- Virtanen, Pauli et al. (2020). “SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python”. In: *Nature Methods* 17, pp. 261–272. DOI: [10.1038/s41592-019-0686-2](https://doi.org/10.1038/s41592-019-0686-2).
- Wooldridge, Lydia K and Beth L Dumont (Jan. 2023). “Rapid Evolution of the Fine-scale Recombination Landscape in Wild House Mouse (*Mus musculus*) Populations”. In: *Molecular*

Biology and Evolution 40.1, msac267. ISSN: 1537-1719. DOI: [10.1093/molbev/msac267](https://doi.org/10.1093/molbev/msac267).
URL: <https://doi.org/10.1093/molbev/msac267> (visited on 09/03/2025).

7 Declaration

Erklärung / Declaration

Hiermit erkläre ich, dass ich die vorliegende Arbeit selbständig und ohne fremde Hilfe angefertigt und keine anderen als die angegebenen Quellen und Hilfsmittel verwendet habe. Die eingereichte schriftliche Fassung der Arbeit entspricht der auf dem elektronischen Speichermedium.

I hereby declare that I have prepared this thesis independently and without outside assistance. I have not used any sources or aids other than those indicated. The submitted written version of the thesis corresponds to the one on the electronic storage medium.

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